



Patrick Höhn

Overview of OpenFOAM GUIs

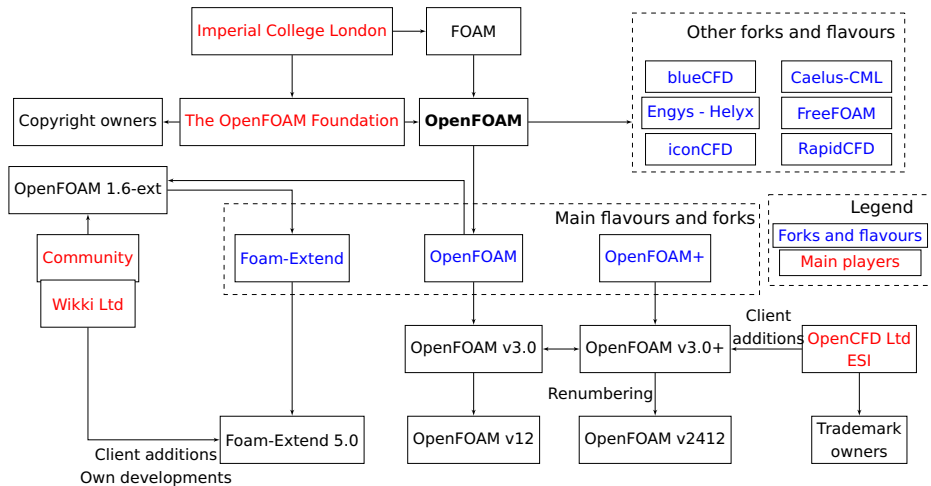
Table of contents

- 1 Introduction
- 2 GUIs for Meshing
- 3 GUIs for complete workflow
- 4 Wrap up

Movitation

- Step learning curve when beginning with OpenFOAM
- Many users not familiar with usual terminal based workflows in OpenFOAM
- Easier editing of meshes
- Competition with commercial products

OpenFOAM



Inspired from: <https://olaflow.github.io/blog/what-is-what-and-who-is-who-in-the-openfoam-environment/>

Discussed GUI Options

■ Meshing

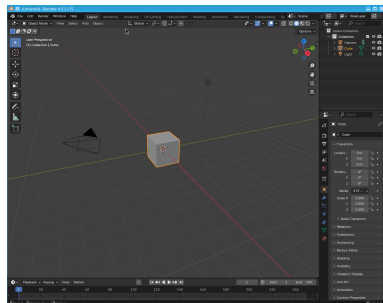
- ▶ Blender
- ▶ SALOME
- ▶ gmsh

■ Complete Workflow

- ▶ Open Source
 - baram
 - insightCAE
 - FreeCAD CFDof and CFD addons
 - Splash
- ▶ Commercial
 - FEATool Multiphysics
 - SimFlow
 - SimWorks
 - Manituflow
 - HELYX/HELXY-OS

Blender

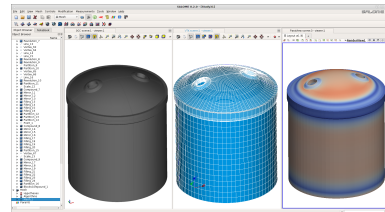
- Open Source 3D animation software
- Available for Windows, Linux and MacOS
- Several available plugins for OpenFOAM Meshing
 - ▶ **swiftBlock** for blockMesh
 - ▶ **venturial** mesh module for snappyHexMesh
 - ▶ **SnappyHexMesh GUI Addon for Blender**



SALOME

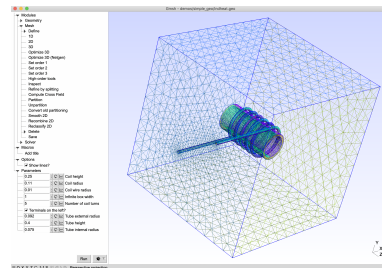
- Open Source Pre- and Post Processing Tool
- Available for Windows and Linux
- More focused on code_saturne and Code_Aster
- Based on OpenCASCADE kernel
- Geometry and Mesh can be generated and exported to OpenFOAM

SALOME



gmsh

- Open Source Finite Element Mesh generator
- Available for Windows, Linux and MacOS
- Based on OpenCASCADE kernel
- Integrated in FreeCAD and FEATool Multiphysics
- Compatible with the FEMCode GetDP
- Mesh can be imported in OpenFOAM



baramCFD

- Available Platforms: Ubuntu 20.04 or later, CentOS 8.2 or alternatives (Rocky Linux, AlmaLinux, ...), OpenSUSE Leap 15.4, Linux Mint 21 "Vanessa", Windows 10 or later, macOS 10.14 or later
- Compatible with OF Versions: Fork of ESI versions
- License: GPLv3
- Current Status:
 - ▶ Unmaintained
 - ▶ Not Compatible with python 3.13
 - ▶ Installation manual tricky to find
 - ▶ Requires code modifications to run, mostly now available on cloud
- Link: <https://github.com/nextfoam/baram>

insightCAE

- Available Platforms: Linux, Windows
- Compatible with OF Versions: openfoam.org, openfoam.com, foam-extend
- License: GPLv2
- Current Status:
 - ▶ Not compatible with latest OpenFOAM versions
 - ▶ Actively maintained
 - ▶ Allows definition of simulation workflows
- Link: <https://github.com/hkroeger/insightcae>

FreeCAD CFDof

- Available Platforms: Windows, Linux, (MacOS)
- Compatible with OF Versions: openfoam.org, openfoam.com
- License: LGPLv3
- Current Status:
 - ▶ Integrated with FreeCAD
 - ▶ Easy Geometry generation
 - ▶ Actively maintained
 - ▶ Covers all steps of simulation setup
- Link: <https://github.com/jaheyne/Cfd0F>

FreeCAD CFD

- Available Platforms: Windows, Linux, (MacOS)
- Compatible with OF Versions: openfoam.org, openfoam.com
- License: GPL
- Current Status: Unmaintained
 - ▶ Integrated with FreeCAD
 - ▶ Unmaintained
 - ▶ Adds FEATool compared to CFDof
- Link: <https://github.com/qingfengxia/Cfd>

Splash

- Available Platforms: Ubuntu
- Compatible with OF Versions: openfoam.org 8-12, openfoam.com 2212-2406
- License: GPLv2
- Current Status:
 - ▶ Maintained
 - ▶ Potential to run on a remote cluster using CloudHPC
 - ▶ Still in development
 - ▶ Integrates Gmesh, ParaView, FreeCAD, Blender
- Link: <https://github.com/cfddose/Splash>

FEATool Multiphysics

- Available Platforms: Standalone: Windows, Linux; MATLAB AddOn: Windows, Linux, MacOS
- Compatible with OF Versions: ESI versions
- License: Commercial
- Current Status:
 - ▶ Requires paid license for non-personal or non-trial usage
 - ▶ Actively maintained
 - ▶ Source code available on GitHub
- Link: <https://www.featool.com>

SimFlow

- Available Platforms: Windows, Linux
- Compatible with OF Versions: OpenFOAM 2212
- License: Commercial
- Current Status:
 - ▶ 30 days trail version available
- Link: <https://sim-flow.com>

SimWorks

- Available Platforms: Windows
- Compatible with OF Versions: OpenFOAM 2112
- License: Commercial
- Current Status:
 - ▶ Professional looking GUI
 - ▶ Limited free version available
 - ▶ Windows only
- Link:
<https://www.idealsimulations.com/simworks-free-cfd-software/>

Manitumflow

- Available Platforms: Windows, Linux
- Compatible with OF Versions: OpenFOAM® 3.0.1, v3.0+, 4.0, 4.1, 5.0, v1606+, v1612+, 1706 or 1712
- License: Commercial
- Current Status:
 - ▶ Demo Version available
 - ▶ Mesh generation based on STL files
 - ▶ No current OpenFOAM Version supported
- Link: <https://mantiumflow.com>

Helyx/Helyx-OS

- Available Platforms: : Linux/Windows, X86/ARM/PowerPC
- Compatible with OF Versions: in-house fork
- License: Mixed. HELYX-core (GPL), HELYX-GUI (Commercial)
- Current Status:
 - ▶ Wide application in Formula 1
 - ▶ Current Version 4.4
 - ▶ Allows remote execution on cluster from Windows machine
 - ▶ Deprecated outdated Open Source Version of GUI available on GitHub
- Link: <https://engys.com/helyx/>, <https://github.com/ENGYS/HELYX-OS>

Wrap up

- Only a selected number of options shown
- Many options do not support latest OpenFOAM versions
- Often limited solver support
- FreeCAD CDFof and Splash seem most promising

Questions?